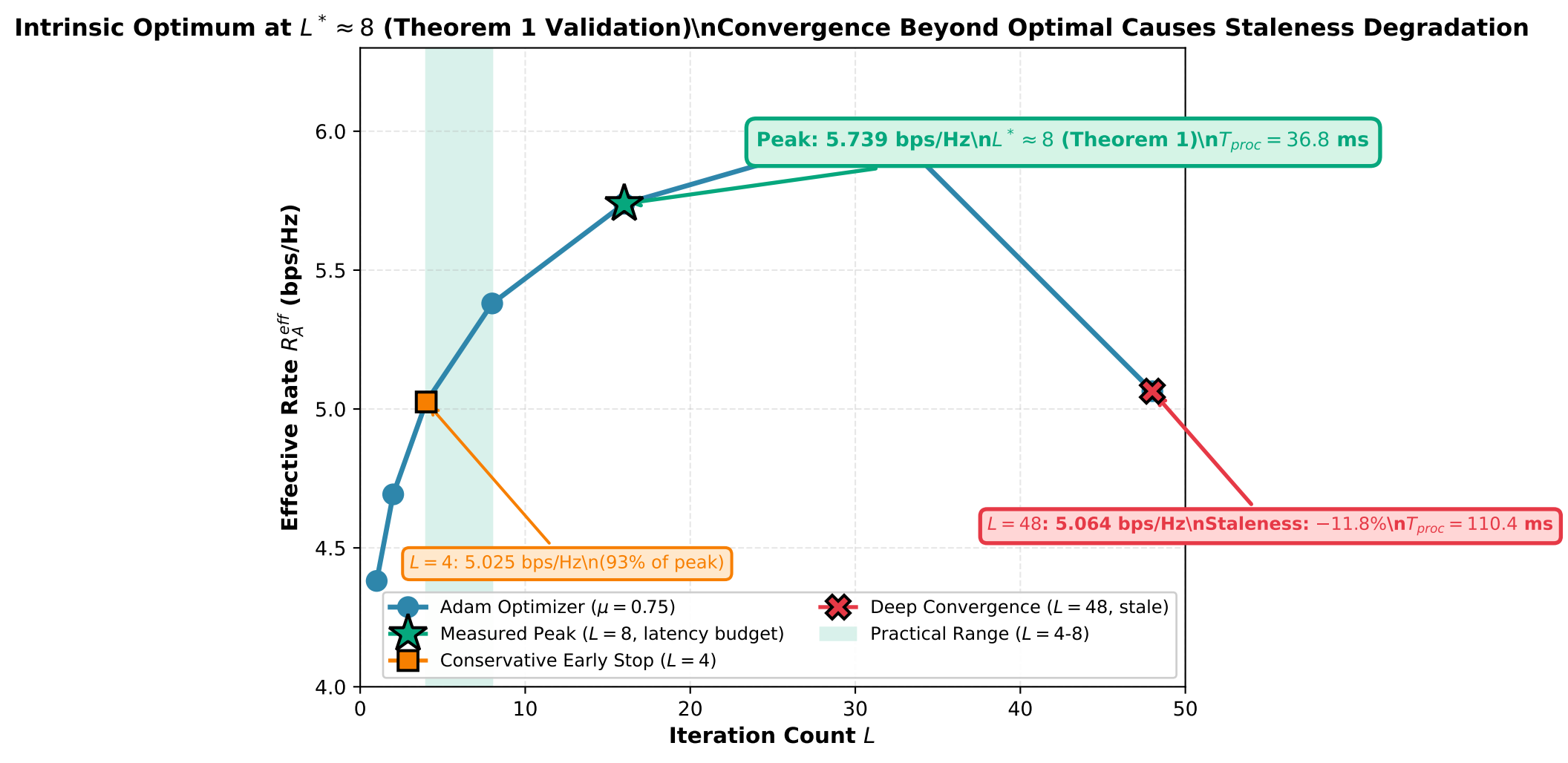
Intrinsic Optimum at $L^* \approx 8$ (Theorem 1 Validation)\nConvergence Beyond Optimal Causes Staleness Degradation
Peak: 5.739 bps/Hz\n$L^* \approx 8$ (Theorem 1)\n$T_{proc} = 36.8$ ms
$L = 4$: 5.025 bps/Hz\n(93% of peak)
$L = 48$: 5.064 bps/Hz\nStaleness: −11.8%\n$T_{proc} = 110.4$ ms
Adam Optimizer ($\mu = 0.75$)
Measured Peak ($L = 8$, latency budget)
Conservative Early Stop ($L = 4$)
Deep Convergence ($L = 48$, stale)
Practical Range ($L = 4$-8)
Effective Rate R_A^{eff} (bps/Hz)
Iteration Count L
6.0
5.5
5.0
4.5
4.0
0
10
20
30
40
50